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User Interface Design

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CSE

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User Interface Design

- User Interface Design Activities
- GUI Implementation Tools
- Command line interface (CLI)

User Interface Design Activities

- There are a number of activities performed for designing user interface. The process of GUI design and implementation is alike SDLC. Any model can be used for GUI implementation among Waterfall, Iterative or Spiral Model.

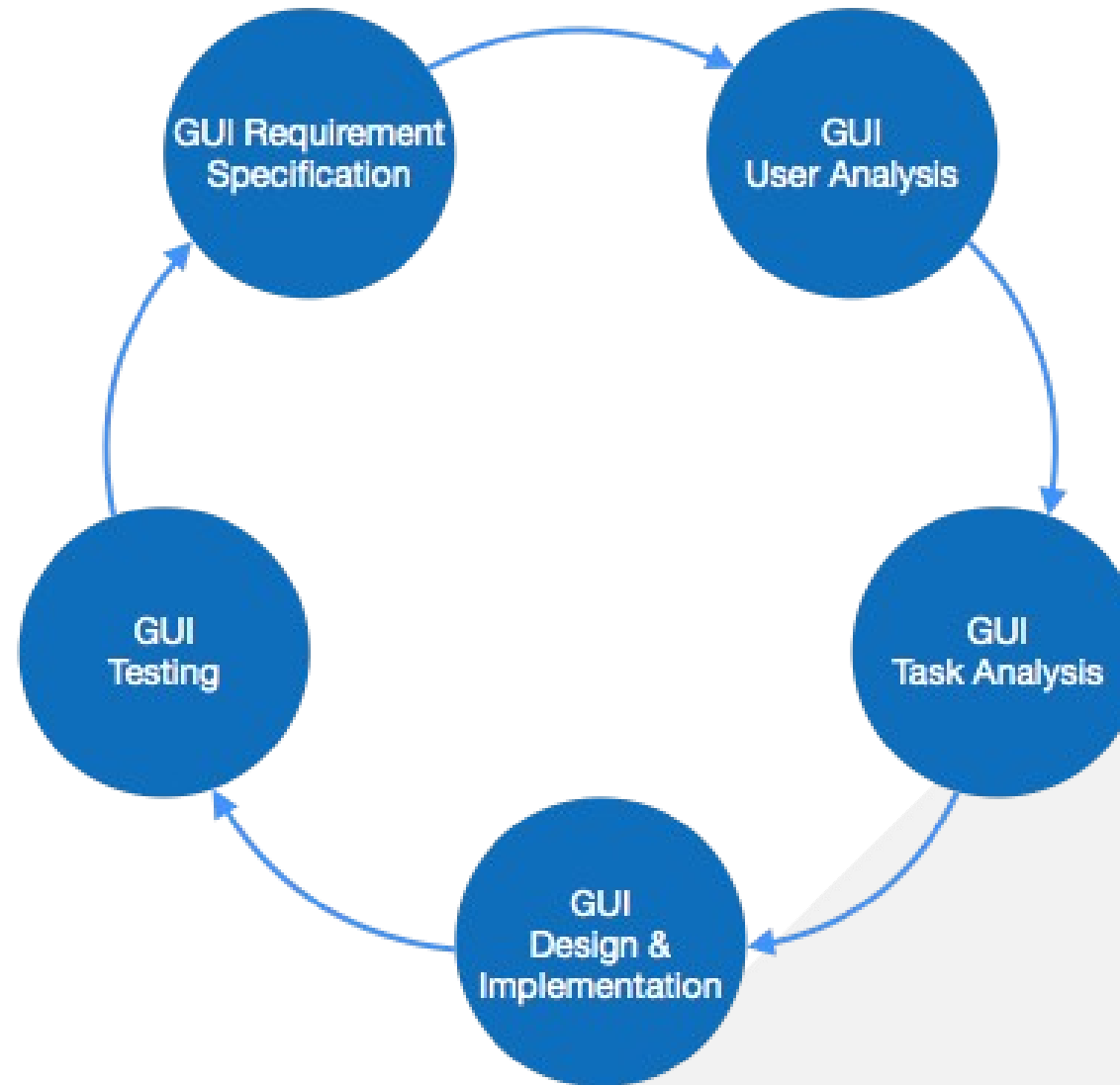
User Interface Design Activities

A model used for GUI design and development should fulfill these GUI specific steps.

- **GUI Requirement Gathering** - The designers may like to have list of all functional and non-functional requirements of GUI. This can be taken from user and their existing software solution.
- **User Analysis** - The designer studies who is going to use the software GUI. The target audience matters as the design details change according to the knowledge and competency level of the user. If user is technical savvy, advanced and complex GUI can be incorporated. For a novice user, more information is included on how-to of software.

User Interface Design Activities

- **Task Analysis** - Designers have to analyze what task is to be done by the software solution. Here in GUI, it does not matter how it will be done. Tasks can be represented in hierarchical manner taking one major task and dividing it further into smaller sub-tasks. Tasks provide goals for GUI presentation. Flow of information among sub-tasks determines the flow of GUI contents in the software.
- **GUI Design & implementation** - Designers after having information about requirements, tasks and user environment, design the GUI and implements into code and embed the GUI with working or dummy software in the background. It is then self-tested by the developers.
- **Testing** - GUI testing can be done in various ways. Organization can have in-house inspection, direct involvement of users and release of beta version are few of them. Testing may include usability, compatibility, user acceptance etc.



GUI Implementation Tools

There are several tools available using which the designers can create entire GUI on a mouse click. Some tools can be embedded into the software environment (IDE).

- GUI implementation tools provide powerful array of GUI controls. For software customization, designers can change the code accordingly.
- There are different segments of GUI tools according to their different use and platform.

Example

- Mobile GUI, Computer GUI, Touch-Screen GUI etc. Here is a list of few tools which come handy to build GUI:
- FLUID
- AppInventor (Android)
- LucidChart
- Wavemaker
- Visual Studio

User Interface Golden rules

The following rules are mentioned to be the golden rules for GUI design

- **Strive for consistency** - Consistent sequences of actions should be required in similar situations. Identical terminology should be used in prompts, menus, and help screens. Consistent commands should be employed throughout.
- **Enable frequent users to use short-cuts** - The user's desire to reduce the number of interactions increases with the frequency of use. Abbreviations, function keys, hidden commands, and macro facilities are very helpful to an expert user.
- **Offer informative feedback** - For every operator action, there should be some system feedback. For frequent and minor actions, the response must be modest, while for infrequent and major actions, the response must be more substantial.

User Interface Golden rules

- **Design dialog to yield closure** - Sequences of actions should be organized into groups with a beginning, middle, and end. The informative feedback at the completion of a group of actions gives the operators the satisfaction of accomplishment, a sense of relief, the signal to drop contingency plans and options from their minds, and this indicates that the way ahead is clear to prepare for the next group of actions.
- **Offer simple error handling** - As much as possible, design the system so the user will not make a serious error. If an error is made, the system should be able to detect it and offer simple, comprehensible mechanisms for handling the error.

User Interface Golden rules

- **Permit easy reversal of actions** - This feature relieves anxiety, since the user knows that errors can be undone. Easy reversal of actions encourages exploration of unfamiliar options. The units of reversibility may be a single action, a data entry, or a complete group of actions.
- **Support internal locus of control** - Experienced operators strongly desire the sense that they are in charge of the system and that the system responds to their actions. Design the system to make users the initiators of actions rather than the responders.
- **Reduce short-term memory load** - The limitation of human information processing in short-term memory requires the displays to be kept simple, multiple page displays be consolidated, window-motion frequency be reduced, and sufficient training time be allotted for codes, mnemonics, and sequences of actions.

Command line interface (CLI)

- A command line interface (CLI) is a text-based user interface (UI) used to view and manage computer files.
- Command line interfaces are also called command-line user interfaces, console user interfaces and character user interfaces.

Prompts and commands

- There are hundreds of different commands available in a command line. The set of commands may vary dramatically between operating systems or applications.
- The following is a list of commands in the Microsoft task configuration and automation framework PowerShell:
- **Get-Date**: Retrieves the current time and date.
- **cd**: Used to change directories.
- **Stop-Process**: Terminates one or more system processes.

Prompts and commands

- Although some commands operate alone, others require the use of arguments. The argument follows the command and provides additional details or specifics.
- For example, the `cd` command tells the OS to change to a different directory, but the command line must include the name of the desired directory path, as well.
- For example, a full command with an argument appears as:
- `cd \documents\user`

CLI advantages and disadvantages

- The advantages of a command line interface are:
- granular control of an OS or application;
- faster management of a large number of operating systems;
- ability to store scripts to automate regular tasks; and
- basic command line interface knowledge to help with troubleshooting, such as network connection issues.

Disadvantages

- The disadvantages of a command line interface are:
- GUI is more user-friendly;
- steeper learning curve associated with memorizing commands and complex syntax/arguments; and
- different commands used in different shells.

CLI versus GUI

- The graphical user interface is the most popular user interface today. A GUI uses windows, menus and icons to execute commands.
- A mouse is the most common way to navigate through a GUI, although many GUIs allow navigation and execution via a keyboard.
- One example of a GUI-based application is Microsoft Word.
- A user can change options for page layouts and styles by selecting the corresponding icon with a mouse or keyboard.

CLI versus GUI

- Administrators who manage thousands of systems or user configurations will find a GUI far less efficient than a CLI.
- But a simple CLI command can easily adjust configurations for a large group of systems at once.
- Commands and arguments can also be combined and saved, then executed as a script each time that specific action -- or comprehensive set of actions -- is required.
- The CLI is the preferred tool for many enterprise-wide systems management tasks.

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THANK YOU